

Lecture 1- Introduction to Data Mining

⇒ What is data?

↳ Data is a collection of raw facts, numbers, text, images, videos, or observations that by themselves do not provide any meaning

Information → when data is processed and organized to answer a question it becomes information.

knowledge → knowledge is obtained when information is analyzed and used for decision-making.

example → students scoring less than 70 mark need extra class

⇒ What is data mining?

It is a process of discovering useful, hidden, and meaningful patterns from a large amount of data. Instead of reading millions of records manually, computers automatically find useful information.

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for example -> Amazon recommendation of products bought together.

=> Why do we need data mining?

every organization stores huge amount of data
example banking, hospital, amazon etc.

The problem is having data is not enough
organization need useful information. This
is why data mining is required.

for example

Banking -> predict loan default

Healthcare -> predict diseases

shopping web -> recommend products.

=> Why data mining become important

Earlier small amount of data, people could
analyze manually, Now huge amount of
data is generated every second. manual
analysis is impossible. Therefore, data
mining is necessary.

⇒ Evolution of Database technology

- Stage 1 ⇒ file system, only files were stored
problem → duplicate data, no security, no sharing
- stage 2 → Database management system [DBMS]
centralized storage, easy retrieval, security, backup.
- stage 3 → Data warehouse → store historical data, supports analysis
- stage 4 → Data mining → finds hidden patterns from data, make predictions, support decision making.

⇒ What is KDD → Knowledge discovery in database
It is a complete process, data mining is only one step inside KDD.

⇒ KDD process

- step 1 → Data collection, collect data from database, excel, internet, sensors etc.

- step 2 → data cleaning, remove missing values, duplicate records, wrong entries.
- step 3 → data integration → combine multiple databases.
- step 4 → data selection, choose only useful data
- step 5 → data transformation, convert data into suitable format.
- step 6 → data mining → apply algorithms, find patterns, predict results.
- step 7 → Pattern evaluation → check pattern useful, is it meaningful
- step 8 → knowledge presentation → show result using graphs, charts, reports, dashboard.

⇒ Application of data mining

(a) Banking → fraud detection, loan approval, cc analysis

- (b) Healthcare → disease prediction, diagnosis
- (c) Education → student performance, attendance analysis
- (d) E-commerce → recommendation system, sale prediction
- (e) social media → friend suggestion, advertisement
- (f) manufacturing → quality control, machine failure
- (g) agriculture → crop prediction, weather analysis
- (h) cyber security → spam detection, virus detection

⇒ Major issues in data mining

- (a) Mining methodology → finding new knowledge
- (b) user interaction → easy interface, visualization
- (c) Efficiency → algorithm should work fast
- (d) Scalability → should handle millions of records
- (e) Different type of data → image, video, text etc.
- (f) Privacy → protect user information, never misuse

⇒ Data Mining Tasks

- (a) Predictive → purpose, predict future
common algorithm → classification, regression, prediction.
- (b) Descriptive → understand existing data

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⇒ Core data mining tasks

- (a) Classification → assign predefined class
- (b) Clustering → create groups, no predefined task
- (c) Association → find relationship
- (d) Prediction → predict future values

⇒ Goals of data mining

- (a) Prediction → forecast future
- (b) Pattern discovery → find hidden relationship
- (c) classification → categorize objects
- (d) clustering → group similar objects
- (e) anomaly detection → find unusual behaviour
- (f) data reduction → reduce huge data, store info
- (g) Knowledge discovery → convert data → knowledge

Lecture 2 -> Main functions of Data Mining

=> What are data mining functions

Data mining functions are the different operations performed on data to discover useful information, hidden patterns, relationships & predictions. They help organizations solve different types of business & research problem

=> Two main categories

(a) Descriptive functions	(b) Predictive function
describe existing data	predict future
"What is happening"	"What will happen"

=> Main function of data mining

(a) Generalization -> It is a process of summarizing detailed data into higher-level useful information.

Benefit -> less data, easy understanding, quick decision making.

(B) Association analysis $\rightarrow$ discover items that frequently occur together

support $\rightarrow$ how frequently

confidence $\rightarrow$ how strongly

(C) classification $\rightarrow$ is the process of assigning data into predefined classes using training data. already defined classes

$\rightarrow$ methods $\rightarrow$

(i) Decision trees

(ii) Naïve Bayes

(iii) Support Vector Machine

(iv) Neural Network

(v) KNN

(vi) Logistic Regression

$\rightarrow$ Application

(i) fraud detection

(ii) medical diagnosis

(iii) email filtering

(iv) disease prediction

(v) customer classification

Bonus

$\rightarrow$ classification
predict classes

$\rightarrow$ prediction $\rightarrow$
numeric value

(d) Cluster Analysis $\rightarrow$ groups similar objects without predefined labels.

maximize similarity inside cluster minimize similarity outside cluster

$\rightarrow$ Application

- (i) customer segmentation
- (ii) market research
- (iii) medical research
- (iv) image processing
- (v) document grouping

(e) Outlier Analysis $\rightarrow$ outliers are data values that are very different from the remaining data.

$\rightarrow$ Applications

- (i) Fraud detection
- (ii) Network security
- (iii) Medical Diagnosis
- (iv) Cyber security
- (v) insurance fraud

⇒ Data Mining Process

→ step 1 → state the problem

→ step 2 → collect the data

→ step 3 → data preprocessing

(i) remove outliers

(ii) scaling → normalization

(iii) encoding → text to numbers

(iv) feature selection → data segmentation

→ step 4 → Build model, apply algorithms

→ step 5 → Interpret Results

⇒ Major Issues in Data Mining

(a) Different Knowledge needs

(b) Background Knowledge

(c) Mining Query language

(d) Result Presentation

(e) Noisy & Incomplete Data

(f) Pattern Evaluation

(g) Efficiency

Lecture 3: Data Warehouse Models

⇒ DBMS → Database management system is a software that stores, organizes, manages, retrieves, and secures data efficiently. It acts as a bridge between the database and users or applications.

⇒ Functions of DBMS

- (a) Data storage → store large amount of data
- (b) Data Retrieval → retrieve data quickly.
- (c) Data integration → combine multiple sources
- (d) Data consistency → avoid duplicate records
- (e) Data security → only authorized users can access data
- (f) Query processing → SQL retrieves required information.

⇒ Role of DBMS in Data Mining

Data mining cannot work without data, DBMS store organized data and data mining analyze it.

⇒ Components of DBMS

(a) Hardware → physical devices → server, disk

(b) Software → program that manages database

(c) Data → Operational Data
Metadata.

(d) Procedures → rules of using DBMS

(e) Database Access Language → Language used to communicate, mostly SQL

DDL → create, alter, drop

DML → insert, update, delete

(f) People → different user, different access

⇒ Data Warehouse → is a central repository that collects, integrates, store, and manages historical data from multiple sources to support analysis & decision making

⇒ characteristics of Data Warehouse

(a) subject-oriented → organize around a subject

(b) Integrated → data comes from multiple source

(c) time-variant → contains historical data

(d) Non-Volatile → data is not frequently updated

⇒ Architecture

(a) Bottom layer [Data collection layer]

E → extract

T → transform

L → load

(b) Middle layer [OLAP server]

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process the queries & allow multidimensional analysis.

(c) Top layer [Presentation layer]
user interact with the system

⇒ OLAP [Online analytical processing]
It allow users to analyze data from different dimensions quickly for decision-making.

⇒ Data Cube ⇒ The face of cube represent different dimension

Dimension → describe data

Fact table → contains numerical data

Dimension table → describe facts

⇒ Operation of OLAP

(a) Roll-UP → summarize data

(b) Drill-down → detailed data

(c) Slice → select one dimension

(d) Dice $\rightarrow$ multiple values from multiple dimensions

(e) Pivot (Rotate) $\rightarrow$ same data different view

$\Rightarrow$ Advantage of OLAP / Data Warehouse

(a) Fast data analysis

(b) Better decision making

(c) Interactive reporting

(d) Trend discovery

(e) Support Business Intelligence

(f)

$\Rightarrow$ Limitations

(a) Expensive to implement

(b) Complex architecture

(c) large storage requirement

(d) require skilled professionals.

Lecture 4 → Machine learning

⇒ Machine learning is a branch of AI that enables computers to learn from data, identify patterns, and improve their performance without being explicitly programmed.

⇒ AI vs ML vs Data Mining



make machine intelligent, ex: chatbots, robots

ML → enable computers to learn from data

DM → extracting useful information from data

⇒ Types of Machine Learning

(a) Supervised Learning ⇒ trained from labeled data, it means correct answer is already known.

→ Applications

(a) Email spam detection

(b) Disease prediction

(c) Loan Approval

(d) Student Result prediction

(e) Face recognition.

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⇒ Two types

(a) Classification → predict predefined categories

(b) Regression → predict the numbers.

⇒ Algorithms

(a) Linear Regression → predict numerical values

(b) Logistic Regression → predict Yes/No.

(c) Decision tree → looks like a flowchart

(d) Support Vector Machine → find the best boundaries

(e) KNN → looks at nearby similar examples.

(f) Naïve Bayes → probability based classification

(g) Random forest → uses many Decision Trees together

(h) Ensemble Learning → combine multiple models to build a strong one

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(a) Bagging $\rightarrow$ train many models independently in parallel, each on random sample of data. the combine their outputs by voting [class] or averaging [regression].

goal $\rightarrow$ reduce over fitting $\Rightarrow$ perform well on training data but perform poor on unseen data. [memorize noise]

(b) Boosting $\rightarrow$ train many models sequentially where each new models try to fix mistake of previous one.

Goal $\rightarrow$ Reduce under fitting $\Rightarrow$ perform poor on training as well as unseen data [not able to understand pattern]

(b) Unsupervised Learning $\Rightarrow$ works with unlabeled data, no correct answer is provided, the computer discover by itself.

$\Rightarrow$ Types of unsupervised learning

(a) Clustering $\rightarrow$ group similar objects

(b) Associate mining rule $\rightarrow$ find relationship

(c) Dimensionality Reduction $\rightarrow$ reduce the number of features while preserving important information.

(c) Reinforcement learning $\rightarrow$ learns through trial & error using rewards & penalties.

$\Rightarrow$ Components

Agent $\rightarrow$ Decision maker

Environment $\rightarrow$ where agent works

Reward $\rightarrow$ Positive Feedback

Penalty $\rightarrow$ Negative Feedback

$\Rightarrow$ Applications

Robotics

Game playing

Self driving Cars

Industrial Automation

$\Rightarrow$ Role of Machine Learning in Data Mining

(a) Pattern Discovery

(b) Classification

clustering
prediction
anomaly detection
decision support

⇒ Application of Machine Learning

Face unlock, Fraud detection, weather forecast, stock market prediction etc.

⇒ Statistics in Data Mining → machine learning learns patterns, but statistics help us understand, summarize, and validate data.

⇒ Two types of statistics

① Descriptive statistics → summarizes data
example → mean, mode, median, standard deviation, correlation.

② Inferential statistics → draw conclusions from sample data

⇒ Statistical Analysis ⇒ It is a process of collecting, organizing, analyzing, interpreting, and presenting numerical data to identify patterns, relationship, and trends. It deals with numbers.

⇒ Non statistical Analysis ⇒ It is a analysis of non-numerical data, such as text, image, audio, videos etc. to extract useful information.

Chapter - 2

Lecture 5: supervised learning

- ⇒ supervised learning: is a machine learning techniques in which the computer learns from labeled data to predict the correct output for new, unseen data.
- ⇒ A label is a correct answer associated with each record.
- ⇒ Feature ⇒ An input variable or characteristic used for prediction
- ⇒ Target variable ⇒ the output we want to predict is called the Target variable or class label
- ⇒ Applications:
 - (a) Email Spam Detection
 - (b) Medical Diagnosis
 - (c) House Price Prediction
 - (d) Loan Approval
 - (e) Student Result Prediction
 - (f) Face recognition.

⇒ Advantages of supervised Learning

- (a) High prediction accuracy [with label data]
- (b) Easy to measure performance because correct ans.
- (c) suitable for both classification & regression
- (d) Widely used in healthcare, banking, education, e-com.
- (e) helps automate decision-making.

⇒ Limitations

- (a) Require a large amount of labeled data.
- (b) Labeling data can be expensive and time-consuming
- (c) Performance depends on data quality
- (d) May overfit if not trained properly
- (e) Cannot learn patterns outside the training data

⇒ What is data ⇒ raw information collected from the real-world that is used to train a machine learning model.

⇒ Dataset ⇒ A simply collection of many data records.

row → data

column → feature or attribute.

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⇒ A record [instance, example or case] is a single row of dataset

⇒ Attributes is an input variable used by the model to make predictions.

⇒ Steps in supervised learning

(a) Training [learning phase]

(b) Testing [Prediction phase]

⇒ step 1 ⇒ data collection

step 2 ⇒ data preprocessing

[missing values, duplicate records, incorrect values, noise, outliers]

step 3 ⇒ Split the dataset

(a) training, (b) testing

step 4 ⇒ Training phase

↳ learn, make model, observe feature

step 5 ⇒ Testing phase

⇒ we test on unlabeled data and check the accuracy.

step 6 ⇒ prediction.

step 7 ⇒ Evaluation

Classification

↳ Accuracy

↳ Precision

↳ F1 score

↳ Recall

Regression

↳ MAE

↳ MSE

↳ RMSE

↳ R^2 score

⇒ Overfitting ⇒ model memorizes the learning data instead of learning general pattern

⇒ underfitting ⇒ model is too simple and fails to learn important patterns from the training.

⇒ Two types in supervised learning

(a) Classification ⇒ is a supervised learning technique used to predict categories or class labels based on input data

(b) Regression ⇒ used to predict continuous numerical values by learning the relationship between input variable and the output variable

Lecture 6 : Classification

⇒ Characteristics of classification

- (a) Predict ~~changes~~ categories
- (b) uses labeled data
- (c) learns pattern
- (d) predict new data

⇒ Components of classification

Data collection

feature extraction

model training

model evaluation

prediction

model improvement [if prediction is not good]

⇒ Types of classification

- (a) Binary classification ⇒ where data point only belong one of only two possible classes.

(b) multiclass classification $\Rightarrow$ data needs to be classified into more than two classes but each point belongs to only one class.

(c) multi-label classification $\Rightarrow$ one data point to belong to multiple categories at same time.

for example $\Rightarrow$ hobbies $\Rightarrow$ reading, writing, etc.

$\Rightarrow$ classification algorithm $\Rightarrow$

$\Rightarrow$ (a) Linear
create a linear
boundary between
classes

(b) non linear
create non-linear bound
between classes and
capture more complex
relationships

$\Rightarrow$ Logistic Regression
SVM
Single-layer perceptron
SGD classifier

kNN
Kernel SVM
Decision Tree
Random forest
Ada Boost
Bagging
Voting
Extra trees
Multilayer ANN

(a) Logistic Regression is used to predict the probability that an input belongs to a particular class. [like yes/no,]
bcz it uses mathematical equation similar to linear regression to calculate a value.

It converts the value into a probability using the sigmoid function, and then probability is converted into class.

Sigmoid function converts any real no. into a value between 0 and 1, making it suitable for prediction estimation.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$\sigma(z)$ = output, z = linear combination of input feature, e = Euler's constant [2.718]

Threshold [0.5],

(b) Support Vector Machine $\Rightarrow$ that finds the optimal hyperplane to separate data into different classes while maximizing the margin between them.

$\Rightarrow$ Hyperplane $\Rightarrow$ decision boundary that separates classes.

$\Rightarrow$ Margin $\Rightarrow$ is the distance b/w the hyperplane and the nearest data points from both classes.

$\Rightarrow$ support vector $\Rightarrow$ are the closest data points to hyperplane, they help to determine the position of separating boundaries.

$\Rightarrow$ Different type of SVM

(a) Linear SVM

(b) Kernel SVM

(c) Polynomial SVM.

(c) Single-layer Perceptron is that classifies input data into one of two classes using a linear boundary. [used for binary classification], simplest artificial neural network [ANN]

⇒ components of ANN

(a) Input features ⇒ values provided to the model like $[x, x_1, x_2]$

(b) weight ⇒ it represents how important the feature is, $[w_1, w_2, w_3]$

(c) bias (b) ⇒ additional value added to the weighted sum.

(d) weight sum ⇒ multiply input with the weight

$$Z = w_1 x_1 + w_2 x_2 + b$$

(e) Activation function ⇒ it decides the final output.

Random

- (d) Stochastic Gradient Descent [SGD] classifier is an optimization algorithm that updates model parameters using one training example at a time to minimize prediction error.
[update weight after every training]

In this we don't go through all training data. we choose random data train on it and repeat.

Put random weight on features, predict the output, calculate error, repeat...

Types $\Rightarrow$ Batch, stochastic, mini-batch.

$\Rightarrow$ For example $\Rightarrow$ Check all papers and give feedback instead check one paper & give feedback.

$\Rightarrow$ Non-linear classifiers.

Neighbors

- (a) k-Nearest Algorithm [K-NN] that classifies a new data point based on the majority class of its

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K nearest neighbors.

$k \Rightarrow$ no. of neighbors

calculate the distance $\Rightarrow$ Euclidean,
Manhattan distance.

(b) Decision tree $\Rightarrow$ it is one of the easiest machine learning algorithm, that classifies data by asking a sequence of question based on the input features until a final decision is reached.

(c) Random forest $\rightarrow$ that builds multiple decision trees and combines their prediction to improve accuracy.

~~in~~ multiple decision trees

$\Downarrow$

different prediction

$\Downarrow$

majority result.

(d) Ada Boost [Adaptive Boosting] $\Rightarrow$ combines multiple weak classifiers to build a strong & more accurate classifier.
• [one after other]

(e) for example $\Rightarrow$ After every class revision we find where students lack in understanding

(e) Bagging $\rightarrow$ [Bootstrap Aggregating]

(f) voting $\Rightarrow$ different algorithm like linear regression, decision trees give result & then the final decision

Insist of different data it do it on different model

(g) Extra trees $\rightarrow$ similar to Random forest but introduces even more randomness while building the trees.

lead
yourself

$\Rightarrow$

Regression

chapter

① Exploratory Data Analysis [EDA]

It is a process of examining, summarizing, and visualizing a data set to understand its:

structure, distribution, Patterns, Relationships, variations, Missing values, outliers.

The primary purpose of EDA is to understand the data mining or machine learning algorithms.

⇒ EDA helps us analyze and understand data so that meaningful patterns, relationship, and anomalies can be discovered.

② Data quality → Data mining results are only as reliable as the data used to produce them.

If data is → incomplete, incorrect, inconsistent, outdated.

then the results of data mining and decision making may also be unreliable.

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"Quality decisions require quality data".

⇒ Measures of Data

Data quality is multidimensional, meaning that it is evaluated using several different dimensions

- (a) Accuracy → It refers to whether the data values are correct or incorrect
- (b) Completeness → Completeness refers to whether all required data is available and recorded.
- (c) Consistency → Consistency means the data does not contain contradictions or discrepancies
- (d) Timeliness → Timeliness refers to whether data is updated and available when it needed.
- (e) Believability → It refers to how much users trust the data.

(f) Interpretability → refers to how much easily users can understand the data.

⇒ Major tasks in data preprocessing

- (a) Data Cleaning
- (b) Data Integration
- (c) Data Reduction
- (d) Data Transformation
- (e) Data Discretization.

⇒ Data cleaning → is the process of identifying and correcting problems in data.
It involves

- (a) Filling Missing values
- (b) Smoothing Noisy data
- (c) Identifying or removing outliers
- (d) Resolving inconsistencies
- (e) Removing Duplicate records.

⇒ Dirty data → Real world data is often called dirty data because it contains errors and imperfections.

The major types are:->

- (a) Incomplete data -> Some data is missing
- (b) Noisy data -> Random errors or unwanted variation in a measured variable.
- (c) Inconsistent Data -> conflicting values, format, codes, or names.
- (d) Duplicate Data -> same record multiple time
- => Incomplete data -> Data is missing when attributes do not have recorded value.

Cause of Missing Data ->

- (a) Equipment Malfunction
- (b) Inconsistent data
- (c) Data entry Misunderstanding
- (d) Data considered unimportant
- (e) failure to record history
- (f) Data not available
- (g) Disguised missing data. for example -> date of birth -> 1 Jan.

⇒ Handle Missing Data → missing values may sometimes need to be inferred from available information.

Possible approaches

- (a) using default value
- (b) using attribute mean
- (c) using attribute median
- (d) using the most frequent value.
- (e) deleting records when appropriate
- (f) Predicting the value using other data

→ Noisy data → noise is random error or variation in a measured variable.

It may arise from:-

- (a) Data entry problems
- (b) Technology limitations
- (c) Inconsistent naming conventions
- (d) Faulty Data collection instruments
- (e) Data transmission problem

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⇒ Methods of handling Noisy Data →

(a) Binning → binning is a smoothing technique

steps →

- ① sort the data
- ② Divide the data into bins
- ③ Smooth the value inside the bin:

The bins may be created using equal-frequency partitioning. [each bins has same number of records]

Smoothing methods

- (a) by Mean
- (b) by Median
- (c) by Bin Boundaries.

In this we replace every value inside bin by the smoothing value.

(b) Regression →

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(c) Clustering → groups similar data objects together. [by grouping the data points that does not fit are called outliers]

(d) computer + human inspection.

⇒ Data Integration → It combines data from multiple sources into a single coherent data source.

[database, data warehouse, data cubes, files].

→ Schema Integration → combining schema from different data sources.

→ Metadata Integration →

(c) Clustering → groups similar data objects together. [by grouping the data points that does not fit are called outliers]

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